1.4.27

EE25BTECH11011 - Navya Banavathu

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Question

If **a** and **b** are the position vectors of A and B respectively, find the position vector of a point C in BA produced such that $BC = 1.5\ BA$

Solution

Given the position vectors of **A** and **B** are **a** and **b**
Let the position vector of point **C** be **c**
The vector BC is given by:

$$\mathbf{BC} = 1.5BA \tag{1}$$

$$(\mathbf{c} - \mathbf{b}) = 1.5\,(\mathbf{b} - \mathbf{a}) \tag{2}$$

we have,

$$\mathbf{AC} = \mathbf{BC} - \mathbf{BA} \tag{3}$$

$$(\mathbf{c} - \mathbf{a}) = (\mathbf{c} - \mathbf{b}) - (\mathbf{a} - \mathbf{b}) \tag{4}$$

$$(\mathbf{c} - \mathbf{a}) = (1.5 - 1)(\mathbf{a} - \mathbf{b}) \tag{5}$$

$$(\mathbf{c} - \mathbf{a}) = 0.5\,(\mathbf{a} - \mathbf{b}) \tag{6}$$

Thus,

$$(\mathbf{c} - \mathbf{a}) : (\mathbf{b} - \mathbf{c}) = 0.5 : 1.5 = 1 : 3 \tag{7}$$

Therefore, C divides AB externally in the ratio $1:3$

Writing the ratio as $k : 1$ we have

$$k = 3 \tag{8}$$

By the section formula (external division), the position vector of C is

$$\mathbf{c} = \frac{k\mathbf{b} - \mathbf{a}}{k - 1}, \quad (k = 3). \tag{9}$$

$$\mathbf{c} = \frac{3\mathbf{b} - \mathbf{a}}{2}. \tag{10}$$

$$\therefore \quad \mathbf{C} = \tfrac{3}{2}\mathbf{b} - \tfrac{1}{2}\mathbf{a}. \tag{11}$$

The above equation can be written as:

$$\mathbf{c} = \begin{pmatrix} \mathbf{a} & \mathbf{b} \end{pmatrix} \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix} \tag{12}$$

Code (C)

C Program

```c
#include <stdio.h>

int main() {
    float ax, ay, bx, by;
    printf("Enter coordinates of A (ax ay): ");
    scanf("%f %f", &ax, &ay);
    printf("Enter coordinates of B (bx by): ");
    scanf("%f %f", &bx, &by);
    int k = 3;
    float cx = (k*bx - ax) / (k - 1);
    float cy = (k*by - ay) / (k - 1);
    printf("Position vector of C is: (%.2f, %.2f)\n", cx, cy);
    return 0;
}
```

Python Code:Import and Define points

Part 1

```python
import matplotlib.pyplot as plt

# --- Data ---
# Position vectors (points A and B)
A = (2, 3) # Example coordinates of A
B = (5, -1) # Example coordinates of B

# Ratio for external division (C divides AB externally in 3:1)
k = 3
# Section formula for external division:
# C = (k*b - a) / (k - 1)
Cx = (k*B[0] - A[0]) / (k - 1)
Cy = (k*B[1] - A[1]) / (k - 1)
C = (Cx, Cy)
print(f"Position vector of C is: {C}")
```

Python Code: Plotting

Part 2

```
# --- Plotting ---
points = [A, B, C]
x_coords, y_coords = zip(*points)

plt.figure(figsize=(7, 6))

# Plot points
plt.scatter(x_coords, y_coords, color='red', s=80, zorder=5)

# Connect A-B (main line)
plt.plot([A[0], B[0]], [A[1], B[1]], color='blue', label='Line AB
    ')

# Show extended line through A-B-C
plt.plot(x_coords, y_coords, '--g', label='Line with C')
```

Python Code: Finishing Plot

Part 3

```python
# Labels
labels = ['A'+str(A), 'B'+str(B), 'C'+str(C)]
for (x, y), label in zip(points, labels):
    plt.text(x + 0.2, y + 0.2, label, fontsize=10)
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.gca().set_aspect('equal', adjustable='box')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('External Division (C divides AB in ratio 3:1)')
plt.legend()
plt.grid(True)
import os
# Your plotting code above...
plt.savefig("figs/fig1.png", dpi=300, bbox_inches="tight")
plt.show()
```

Graphical Representation

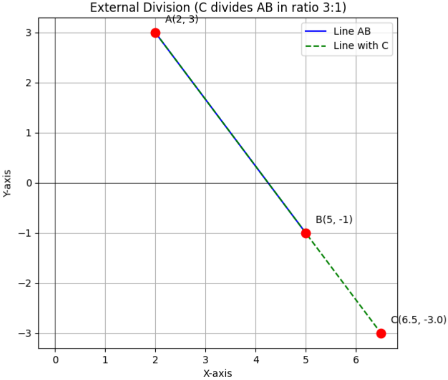

Figure: Point C dividing AB externally in ratio $3:1$